*ALERT meeting*

# AHDC alignment - updates

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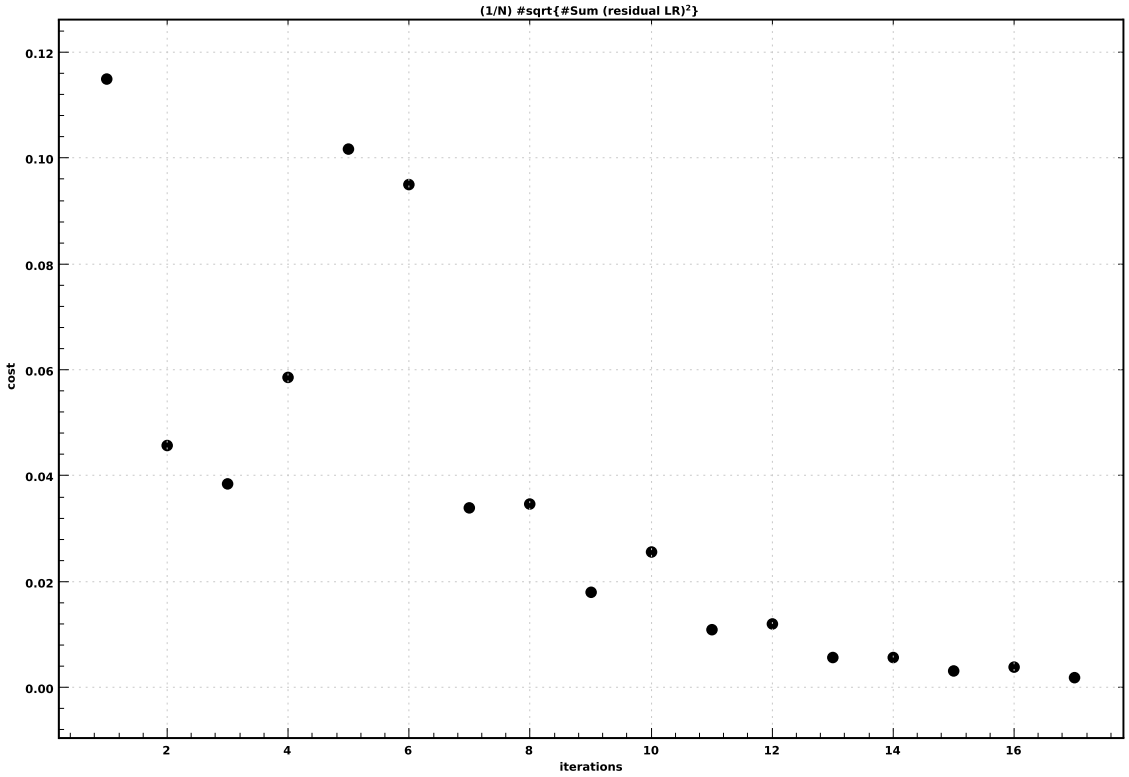
# Updates

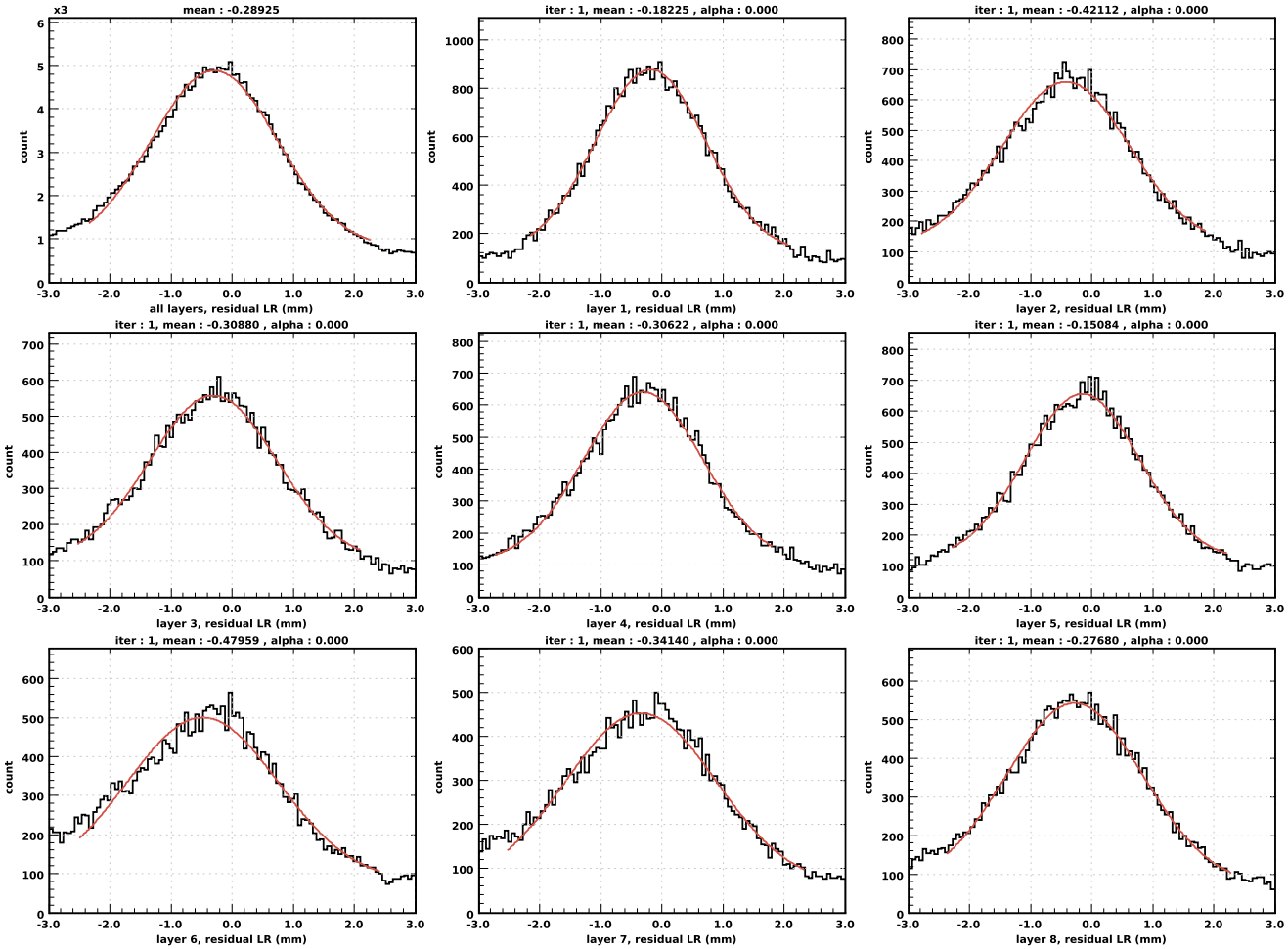
- I mostly worked on the automatisation of the alignment procedure
- For each iteration, I analyse the distribution of the residuals LR and I estimate new rotation angles

$$\text{criteria} = \frac{1}{N}\sqrt{\sum_{\text{layer}} (\text{residual_LR})^2}$$



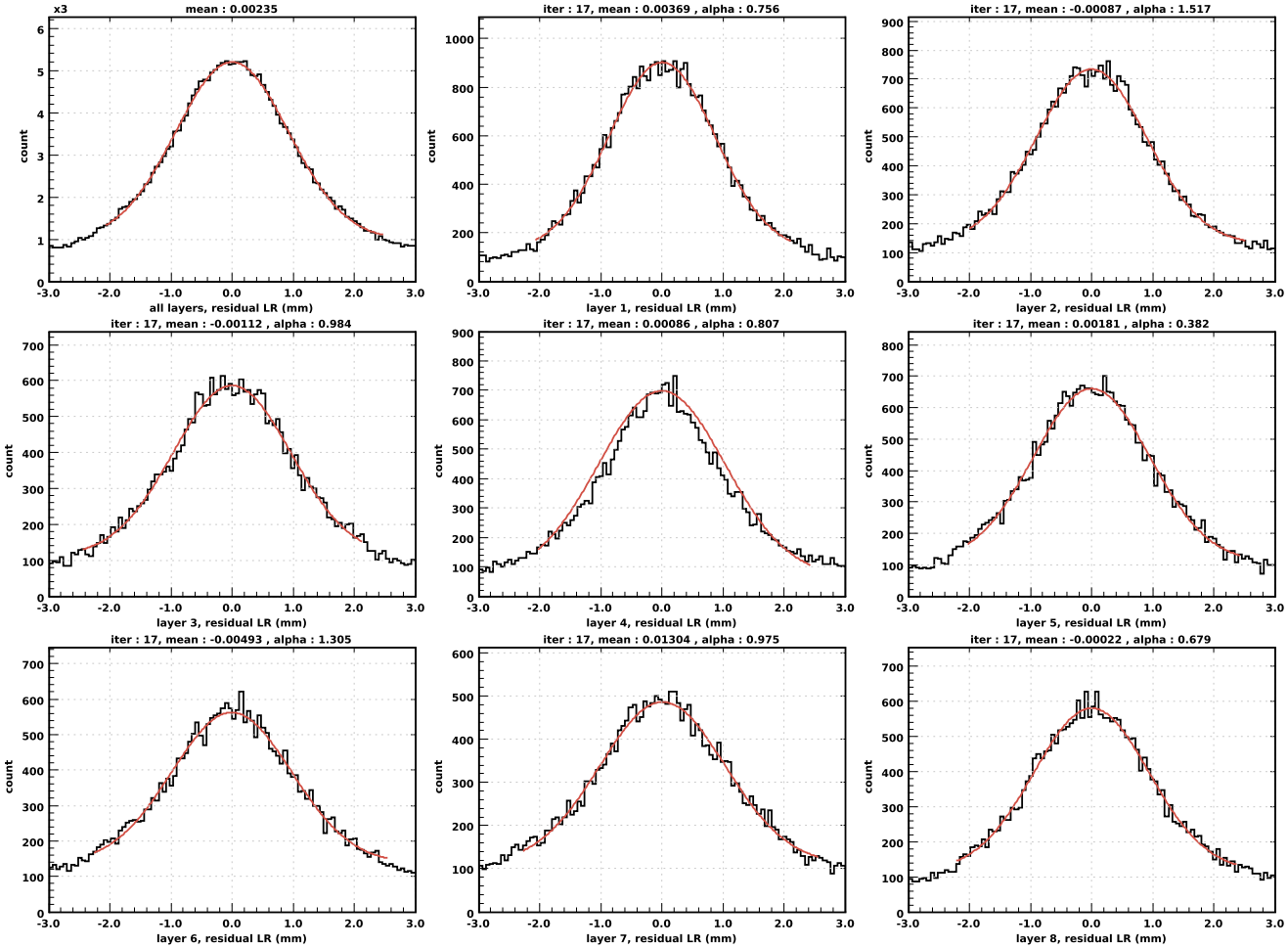
# Updates

- Example: initial distributions per layer (no alignment, $\alpha = 0°$)



# Updates

- Example: final distributions per layer for: criteria $< 2 \cdot 10^{-3}$ mm

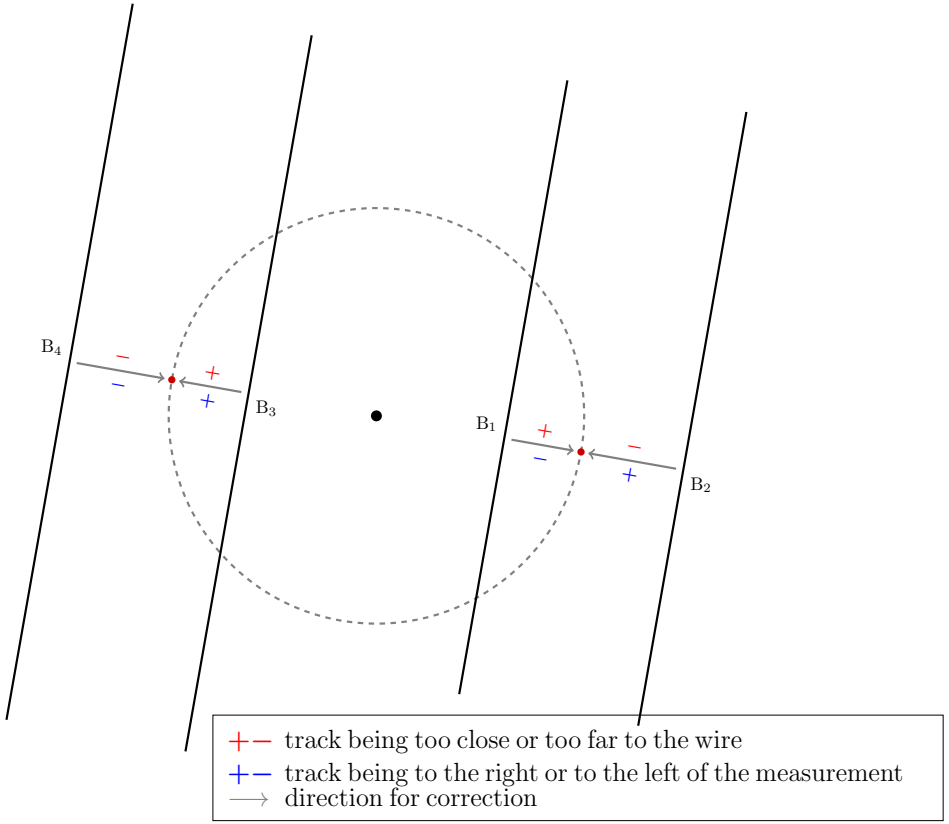


# Observation

- Detector position with respect to CLAS
- The numbers inside the table are the rotation angles in deg

| AHDC position | -54 mm [1] | -30 mm | -70 mm |
|---|---|---|---|
| layer 11 | 0 | -1.49 | +0.756 |
| layer 21 | +2.55 | +3.82 | +1.517 |
| layer 22 | +2.11 | +3.24 | +0.984 |
| layer 31 | 0 | -1.54 | +0.807 |
| layer 32 | -0.32 | -1.96 | +0.382 |
| layer 41 | +3.0 | +3.73 | +1.305 |
| layer 42 | +2.13 | +3.3 | +0.975 |
| layer 51 | -0.21 | -1.61 | +0.679 |

[1] hand made

# Backup slides

- Procedure:
  1. We looked at elastic events : electron + track (here: collection of hits only)
  2. We computed the theoretical momentum vector of the track $\vec{p}$
  3. We made the propagation without doing any fit (i.e no KF correction)
  4. We looked at residual_LR (left-right) $\longrightarrow$ sensitive to the alignment